Fiscal Year:	FY 2016	**Task Last Updated:**	FY 08/21/2015
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Individual Differences in the Neurochemical and Behavioral Response to Exposure to Protons and HZE Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	10/23/2012	**End Date:**	10/22/2016
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 10/22/2016 per PI and NSSC information (Ed., 8/28/15)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Shukitt-Hale, Barbara (US Department of Agriculture/Agricultural Research Service, Beltsville)		
Grant/Contract No.:	NNX13AB73G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Future missions in space (such as a mission to Mars or the establishment of a moon base) will involve long-term travel beyond the magnetic field of the Earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. Cosmic rays are composed of protons, alpha particles, and particles of high energy and charge (HZE particles). Previous research has shown that exposure to HZE particles and protons can affect neural functioning and behavior. These changes may affect the ability of astronauts to successfully meet mission requirements or may produce degenerative changes in neural and behavioral functioning after the conclusion of the mission. The objectives of the proposed research program are: (1) to evaluate the effects of repeated subthreshold exposures to HZE particles and protons on cognitive and neuronal deficits and how they relate to the characteristics of the specific particle; (2) to determine the relationship between changes in performance and HZE particle- and proton-induced changes in oxidative		

	stress, inflammation, signaling molecules, and autophagy in key brain regions; and (3) to determine how individual characteristics, including age and sex, may influence the responsiveness of the individual to HZE particle irradiation. To achieve these objectives, we propose using behavioral and neurochemical measures that have been previously shown to be sensitive to exposure to HZE particles and protons and which may provide a basis for defining the risks of exposure to heavy particles on brain functioning and related behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Two sets of experiments have been performed. The first series of experiments evaluated the effects of repeated sub-threshold exposure to 56Fe. Most research on the cognitive effects of exposure to the types of radiation encountered in space has used a single suprathreshold dose of HZE particles or protons. However, on exploratory class missions it most likely that the total dose received by astronauts will be spread across the duration of the mission. Performance was evaluated on the plus-maze (baseline anxiety), novel spatial recognition (spatial learning and memory), novel object recognition (non-spatial learning and memory), and operant responding (motivation to obtain reinforcement and responsiveness to changes in environmental contingencies). The radiations were single exposures to 15 or 30 cGy, or two exposures to 15 cGy separated by either 24 or 72 hr. The results suggest that the effects of multiple exposures to 56Fe particles are task-dependent. Plus-maze performance showed no effects following single-dose exposure to either 15 or 30 cGy. However, there was a significant increase in the amount of time spent in the anxiogenic open arms of the maze following two exposures to 15 cGy of 56Fe particles separated by 24 or 72 hours. A similar pattern was observed for novel spatial recognition. In contrast, performance on the novel object recognition task and on the operant task was disrupted only by a single dose of 30 cGy. Other than noting that the effects of multiple exposures apparently are task-dependent, the factors that affect responding under these exposures conditions are not known. We have retested the subjects on these tasks to evaluate the effects of aging on performance. As noted with the initial tests, the effects of repeated subthreshold exposures to 56Fe particles varied across the tasks. The performance of the radiated animals did not differ from that of the controls; although that may have reflected the fact that the non-irradiated controls did not spend more time in the novel location. The novel object recognition performance of all subjects exposed to 30 cGy of 56Fe particles (either as a single dose or as two subthreshold doses of 15 cGy) did show a lack of recognition memory. In contrast, exposure to a single dose of either 15 or 30 cGy, but not two separate exposures to 15 cGy, disrupted operant at the higher reinforcement contingencies. Continuing our pilot studies of the effects of repeated sub-threshold exposures to HZE particles, during NSRL-15A, rats were exposed to 8 or 16 cGy of 56Fe particles using the same design that was detailed above. Following irradiation, subsets of the rats were sacrificed for analysis of oxidative stress and neuroinflammation. Neurochemical analyses are currently being conducted at the Human Nutrition Research Center on Aging (HNRCA). The remaining rats were sent to University of Maryland, Baltimore County (UMBC) for cognitive testing. There were no effects on any measure of cognitive performance. They will be re-exposed to 8 or 30 cGy of 56Fe particles during NSRL-15C, and returned to UMBC for further testing. The second series of experiments was designed to evaluate the interactions between age at the time of irradiation, particle linear energy transfer (LET) and the dose needed to produce a disruption of cognitive performance. Previous research had shown that the dose needed to disrupt cognitive performance by exposure to 56Fe particles was lower in older organisms (12-14 months) than in younger subjects (2 months). A similar effect was not observed following exposure to 16O particles. We proposed that the failure to observe an interaction between age and the dose of 16O particles needed to disrupt cognitive performance was related to the fact that the dose needed to disrupt cognitive performance was much lower (˜ 1-5 cGy) than the dose needed to disrupt cognitive performance following exposure to 56Fe particles (˜ 25-30 cGy). This hypothesis was evaluated by exposing 2, 11, and 16 month old rats to 48Ti particles. The results showed that exposing 11 month old, but not 2 month old, rats to 1 or 10 cGy of 48Ti particles disrupted cognitive performance. As such, these results would be consistent with the hypothesis that particle LET is a factor affecting the relationship between age and the responsiveness of the organism to exposure to HZE particles.
Bibliography Type:	Description: (Last Updated: 05/08/2025)
Articles in Peer-reviewed Journals	Poulose SM, Bielinski DF, Carrihill-Knoll KL, Rabin BM, Shukitt-Hale B. "Protective effects of blueberry- and strawberry diets on neuronal stress following exposure to 56Fe particles." Brain Res. 2014 Dec 17;1593:9-18. Epub 2014 Oct 22. http://dx.doi.org/10.1016/j.brainres.2014.10.028 ; PubMed PMID: 25451098 , Dec-2014
Articles in Peer-reviewed Journals	Rabin BM, Heroux NA, Shukitt-Hale B, Carrihill-Knoll KL, Beck Z, Baxter C. "Lack of reliability in the disruption of cognitive performance following exposure to protons." Radiat Environ Biophys. 2015 Aug;54(3):285-95. Epub 2015 May 3. http://dx.doi.org/10.1007/s00411-015-0597-2 ; PubMed PMID: 25935209 , Aug-2015
Articles in Peer-reviewed Journals	Rabin BM, Poulose SM, Carrihill-Knoll KL, Ramirez F, Bielinski DF, Heroux N, Shukitt-Hale B. "Acute effects of exposure to 56Fe and 16O particles on learning and memory." Radiat Res. 2015 Aug;184(2):143-50. Epub 2015 Jul 24. http://dx.doi.org/10.1667/RR13935.1 ; PubMed PMID: 26207687 , Aug-2015
Articles in Peer-reviewed Journals	Rabin BM, Carrihill-Knoll K, Shukitt-Hale B. "Comparison of the effectiveness of exposure to low-LET helium particles (4He) and gamma rays (137Cs) on the disruption of cognitive performance." Radiat Res. 2015 Sep;184(3):266-72. Epub 2015 Aug 18. http://dx.doi.org/10.1667/RR14001.1 ; PubMed PMID: 26284421 , Sep-2015